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Prepare Reversible-Gate Specifications

Gate	Inputs	Outputs	Main Function
NOT	A	Y	$Y = \text{NOT } A$
Feynman / CNOT	A, B	P, Q	$P=A, Q=A \oplus B$
Toffoli	A, B, C	P, Q, R	$P=A, Q=B, R=C \oplus (A \cdot B)$
Fredkin	C, A, B	P, Q, R	Controlled swap of A and B

Symbol

Boolean equations

Truth table

Control input(s)

Target input(s)

Useful outputs

Garbage outputs, if applicable

Constant/ancilla inputs, if required